HW3 – Submitted by Erdelyi, Zsofia & Chu, Tri-Tai

*Hypothesis*

Sobchuk et al. [2] established the positive relationship between early genre entry and subsequent success. We suspect that an artist's/band's position in the artist network (alias their network centrality) is another predictor of success.

*Problem & Research Gap*

Although previous research has established the importance of first-mover advantage in music [1], building on the work of Sagalnik et al., who found that success in artistic domains is not only determined by quality, but also by social information [2], less is known of how the network position is shaping artistic success [3]. In particular, it remains unclear whether artists' positions within collaboration networks influence their success. We suspect that not only does it matter when an artist/band entered a genre, but also who they collaborated with at that time.

*Data for measuring artists' centrality: artist collaboration network*

- Nodes: individual artists
- Links: collaborations between artists (e.g. featured appearances, co-produced songs, shared recordings, shared touring opportunities)
- Possible data sources: setlist.fm (touring), discogs (artists discography)

Why this works: It captures the actual social and professional connections through which artists might gain access to audiences, knowledge and opportunities.

The network centrality measures of the artists provides the initial variables we would like to use. The centrality measures could include degree centrality, eigenvector centrality and betweenness centrality. It should ideally be measured before an artist enters the genre. Otherwise, successful early entry could itself generate collaborations and increase centrality, making it difficult to establish which came first. Based on these centrality measurements we want to quantify the artists' collaboration influence.

*Margin of success*

Our primary possible data source would be librespot/spotipy, which expose Spotify playcount information. Playcounts provide an observable indicator of audience reach and popularity, allowing us to compare the success of artists across genres.

Plan B: youtube playcounts from SocialBlade which is historical YouTube statistics that can provide daily channel-level views, with up to three years of historical data for YouTube. However, these are channel-level rather than necessarily music-specific views.

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*Expected results*

We are investigating how the same opportunity can generate different outcomes depending on an artist's position in a network. If the hypothesis holds, our findings could suggest that:

1. Early entry alone does not guarantee success.
2. Artists with access to influential collaborators may be better positioned to benefit from emerging musical trends.
3. Network structures can reinforce existing inequalities in cultural markets.
4. Cultural innovation may be driven not only by the creation of new genres but also by the social structures through which these innovations spread (knowledge diffusion).

The results could give important insights for emerging artists and other industry professionals like bookers or tour managers, in making more informed choices about collaborations and networking.

*Papers*

[1] Sobchuk et al., "First-mover advantage in music". (https://link.springer.com/article/10.1140/epjds/s13688-024-00476-z)

[2] Sagalnik et al., "Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market". (https://www.science.org/doi/10.1126/science.1121066)

[3] Fraiberger et al., "Quantifying reputation and success in art". (https://www.science.org/doi/10.1126/science.aau7224)